

Clustering of multivariate tail dependence using conditional methods

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**UK Research
and Innovation**



Storm Darragh causes blackouts and travel disruption



Problem

- Often want to estimate

$$\mathbb{P}(X > x, Y > y) = \mathbb{P}(Y > y \mid X > x)\mathbb{P}(X > x)$$

for large x, y

- “concomitant”/concurrent extreme events for random vector $\mathbf{X}$ often particularly devastating
- Goal: **identify trends by clustering sites with similar tail dependence**

Dependence Modelling

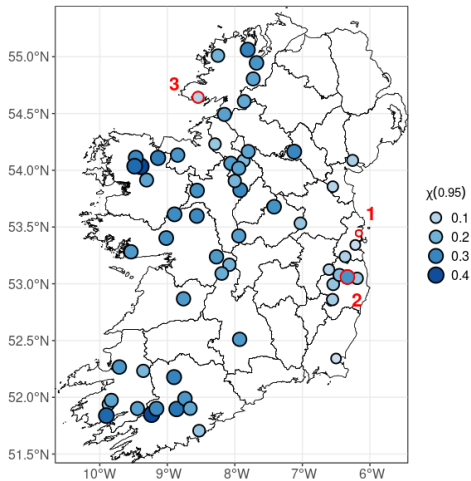
Asymptotic dependence

- Coefficient of extremal dependence $\chi \in [0, 1]$,

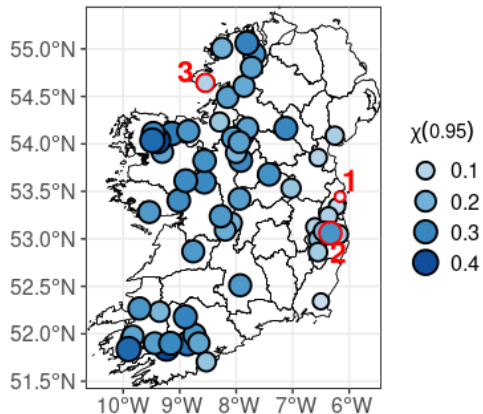
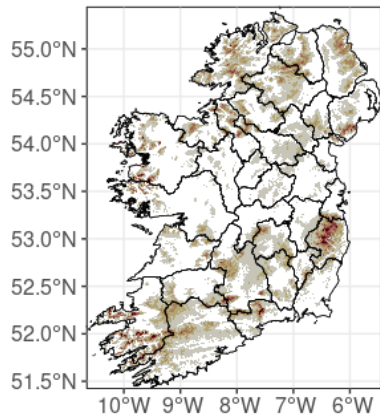
$$\chi = \lim_{u \rightarrow 1} \mathbb{P}[F_1(X_1) > u \mid F_2(X_2) > u]$$

- (Increasingly strong) asymptotic dependence for $\chi > 0$.
- However, χ only gives summary; inference requires **dependence model**.

Ireland

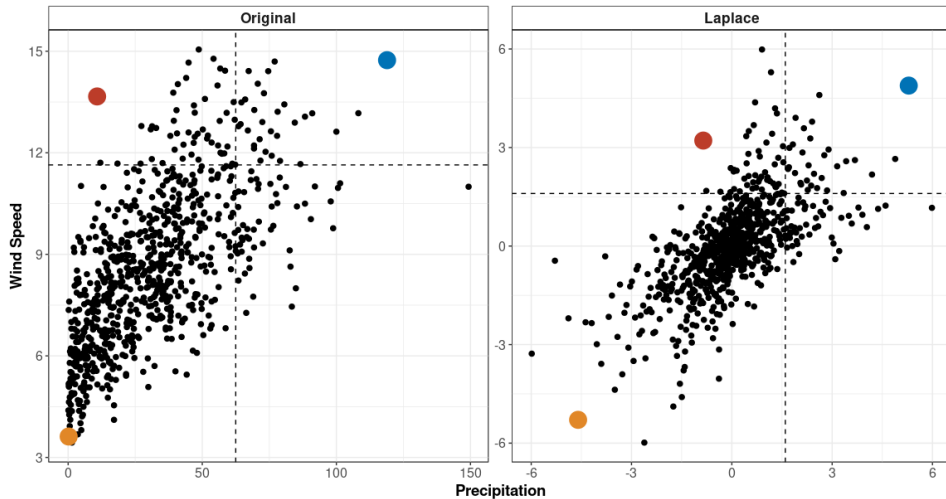


Ireland



Conditional extremes

Marginal transformation



Conditional extremes

- Heteroskedastic regression dependence model:

$$(Y \mid X = x) = \alpha x + x^\beta Z, \text{ for } x > u$$

- slope parameter α for Y given large X ,
- $\beta \in (-\infty, 1]$ controls “spread” induced by large X .

Conditional extremes

- Heteroskedastic regression dependence model:

$$(Y_{-i} \mid Y_i = y) = \alpha_{|i} y + y^{\beta_{|i}} Z_{|i}, \text{ for } y > u_i$$

- slope parameter $\alpha_{j|i} \in [-1, 1]$ for Y_j given large Y_i ,
- $\beta_{j|i} \in (-\infty, 1]$ controls “spread” induced by large Y_i .

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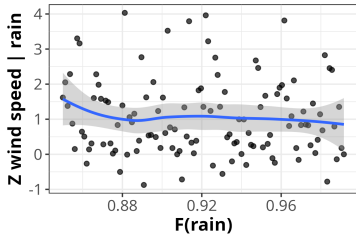
$$(Y_{-i,s} \mid Y_{i,s} = y) = \alpha_{|i,s} y + y^{\beta_{|i,s}} Z_{|i,s}, \text{ for } y > u_{i,s}$$

- slope parameter $\alpha_{j|i,s} \in [-1, 1]$ for $Y_{j,s}$ given large $Y_{i,s}$,
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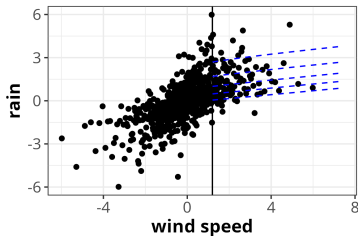
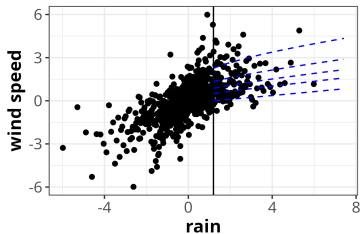
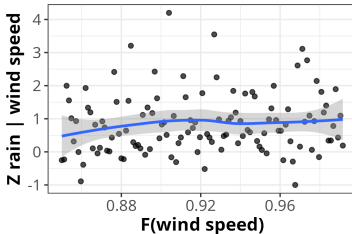
Conditional extremes

Costelloe Fishery

$a = 0.137, b = 0.052$



$a = 0.147, b = 0.243$



Inference

Inference assumes conditional distribution follows a multivariate Normal (MVN) distribution:

$$(\mathbf{Y}_{-i,s} \mid Y_{i,s} = y) \sim \text{MVN}_{d-1} \left(\boldsymbol{\alpha}_{|i,s} y + y^{\boldsymbol{\beta}_{|i,s}} \boldsymbol{\mu}_{|i,s}, (y^{\boldsymbol{\beta}_{|i,s}})^{\text{T}} \boldsymbol{\Sigma}_{|i,s} y^{\boldsymbol{\beta}_{|i,s}} \right),$$

where $y > u_{i,s}$.

$\implies$ dependence structures at different sites can be compared using their MVN distributions

Clustering

Kullback-Leibler divergence

- KL between two continuous $(d - 1)$ -variate distribution functions H and H^* is

$$\text{KL} (H \| H^*) = \int_{\mathbb{R}^{d-1}} h(y) \log \left(\frac{h(y)}{h^*(y)} \right) \mathrm{d}y,$$

where h, h^* are continuous density functions for H, H^*

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- KL has closed form when H and H^* are MVN
- $\text{KL} (H \| H^*) + \text{KL} (H^* \| H)$ symmetric but can be unstable

Skew-geometric Jensen-Shannon divergence

- JS^{G_λ} between H, H^* is

$$\begin{aligned}\text{JS}^{G_\lambda}(H \| H^*) &= (1 - \lambda) \text{KL} \{G_\lambda(H, H^*) \| H\} \\ &\quad + \lambda \text{KL} \{G_\lambda(H, H^*) \| H^*\}.\end{aligned}$$

- $G_\lambda(H, H^*) = H^{1-\lambda}(H^*)^\lambda$ is geometric mean with $\lambda \in [0, 1]$

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- $G_\lambda(H, H^*) = H^{1-\lambda}(H^*)^\lambda$ is geometric mean with $\lambda \in [0, 1]$
- Symmetric, closed form, more stable than KL

Conditional $\text{JS}^{\text{G}_\lambda}$

- $\text{JS}^{\text{G}_\lambda}$ for sites s, s^* *conditional* on variable i :

$$\text{cJS}_{i,s,s^*}^{\text{G}_\lambda}(y) = \text{JS}^{\text{G}_\lambda} (H_{i,s}(y) \| H_{i,s^*}(y)) , \quad y > u_i,$$

where $H_{i,s}(y), H_{i,s^*}(y)$ are conditional MVNs at s, s^*

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where $H_{i,s}(y), H_{i,s^*}(y)$ are conditional MVNs at s, s^*

- $\text{cJS}_i^{\text{G}_\lambda}$ depends on choice of $y > u_i \dots$

Expected $\text{cJS}_{i,s,s^*}^{G_\lambda}$

Expected $\text{cJS}_{i,s,s^*}^{G_\lambda}$:

$$\begin{aligned} \text{eJS}_{i,s,s^*}^{G_\lambda} &= \mathbb{E} \left(\text{cJS}_{i,s,s^*}^{G_\lambda}(Y_{i,s}) \mid Y_{i,s} > u_i \right) \\ &= \int_{u_i}^{\infty} \text{cJS}_{i,s,s^*}^{G_\lambda}(y) h(y \mid Y_{i,s} > u_i) dy \end{aligned}$$

where $h(y \mid Y_{i,s} > u_i)$ is on standard Laplace scale.

Dissimilarity Matrix

- For each conditioning variable $i \in \{1, \dots, d\}$, compute pairwise $\text{eJS}^{\text{G}_\lambda}$ dissimilarities between all D sites:

$$M^{(i)} := \left(\text{eJS}_{i,s,s^*}^{\text{G}_\lambda} \right)_{s,s^*=\{1,\dots,D\}}$$

- Can aggregate over all conditioning variables to get overall dissimilarity matrix:

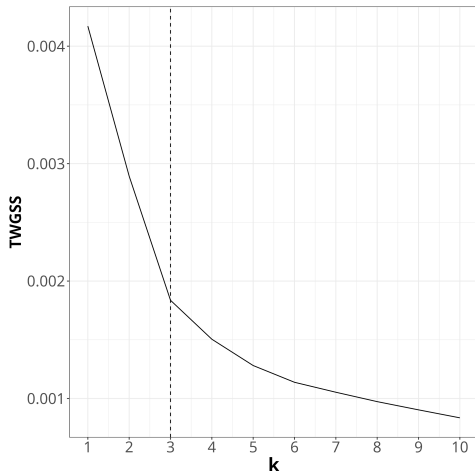
$$M = \frac{1}{d} \sum_{i=1}^d M^{(i)}$$

Clustering

- Use **k-medoids** to cluster over M (or $M^{(i)}$)
- **Elbow plot:** k chosen using **Total Within Group Sum of Squares**:

$$\text{TWGSS}(k) = \sum_{c=1}^k \sum_{s \in \mathcal{C}_c} \text{eJS}_{i,s,\kappa_c}^{\text{G}_\lambda},$$

where $\mathcal{C}_c$, κ_c are cluster c and its medoid

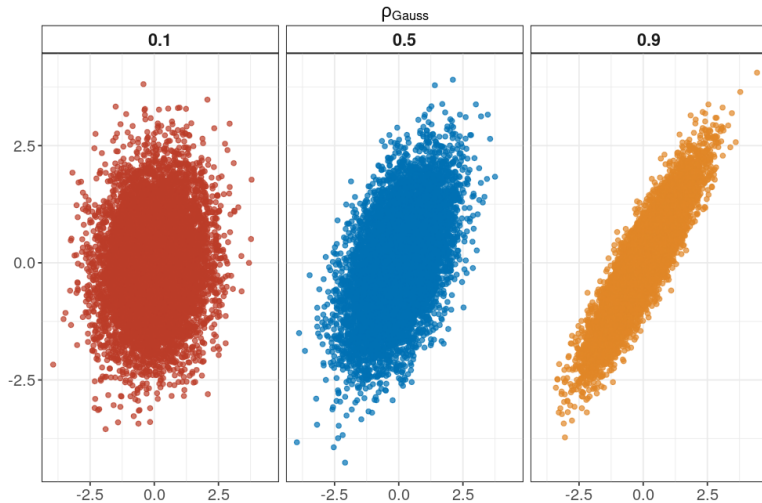


Simulations

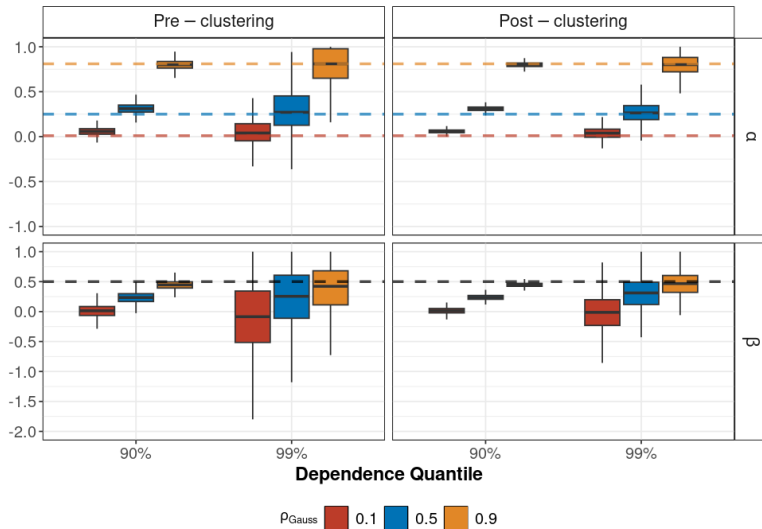
Gaussian copula

- 10,000 observations of $d = 2$ variables at $D = 12$ “sites”
- Bivariate Gaussian copula with correlation $\rho_{\text{Gauss}}^s \in [0, 1]$ at site $s = 1, \dots, D$.
- (asymptotic) CE parameters are $\alpha^s = (\rho_{\text{Gauss}}^s)^2, \beta^s = 1/2$
- Example: 3 clusters of 4 locations defined by respective ρ_{Gauss}^s values of 0.1, 0.5, 0.9.

Gaussian copula



Gaussian copula - results



Mixture simulation

- Mixture of Gaussian and t-copulas, inducing asymptotic independence and dependence, respectively

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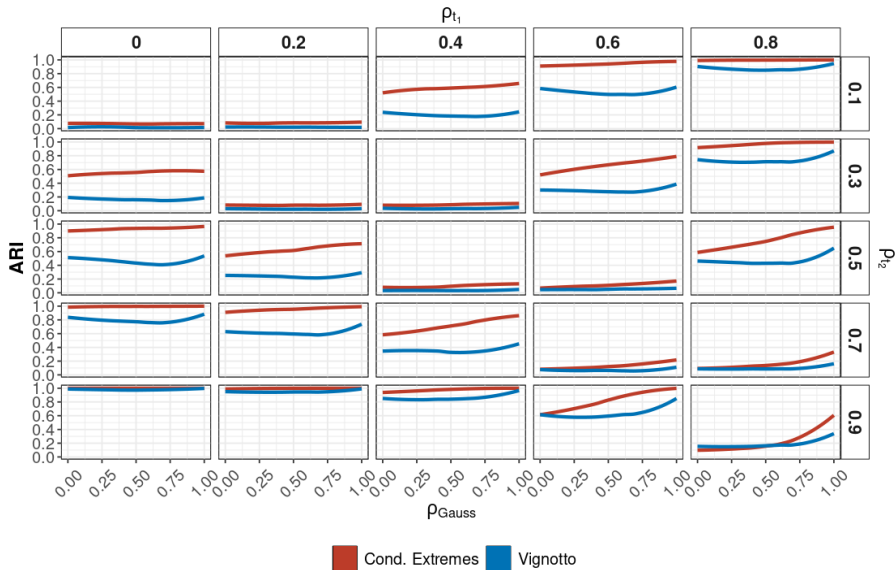
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- Grid search performed over $\rho_{\text{Gauss}}^s, \rho_t^s = \rho_{t_1}, \rho_t^s = \rho_{t_2} \in [0, 1]$.

Mixture simulation

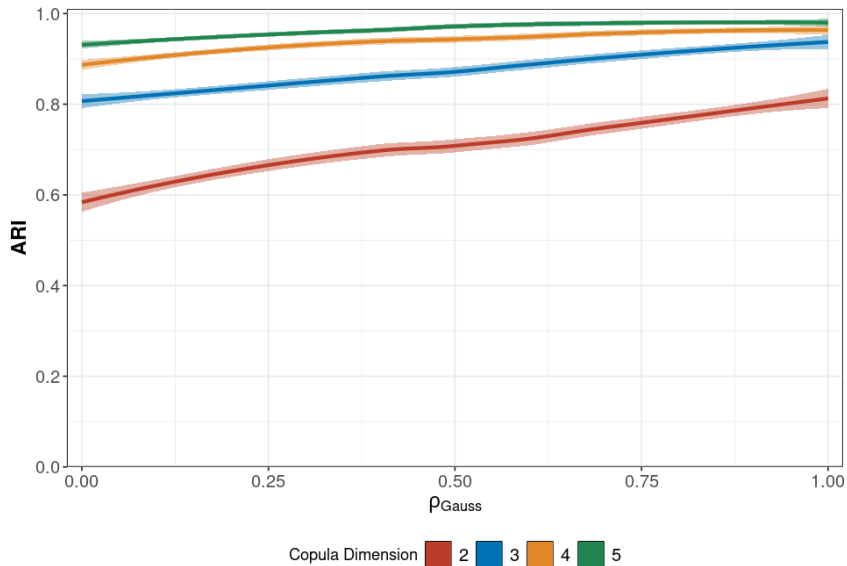
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- Grid search performed over $\rho_{\text{Gauss}}^s, \rho_t^s = \rho_{t_1}, \rho_t^s = \rho_{t_2} \in [0, 1]$.
- Clustering compared to competing Vignotto algorithm using Adjusted Rand Index $\text{ARI} \in [-1, 1]$

Vignotto, Engelke, and Zscheischler,
“Clustering bivariate dependencies of compound precipitation and wind extremes over Great Britain and Ireland” (2021)

Mixture simulation

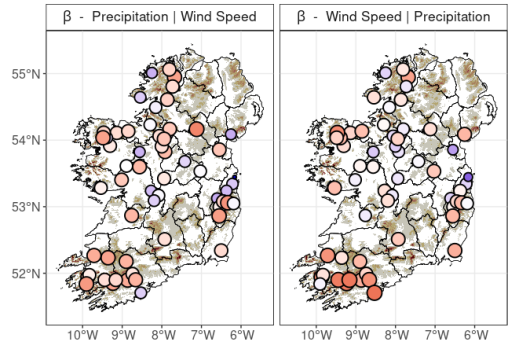
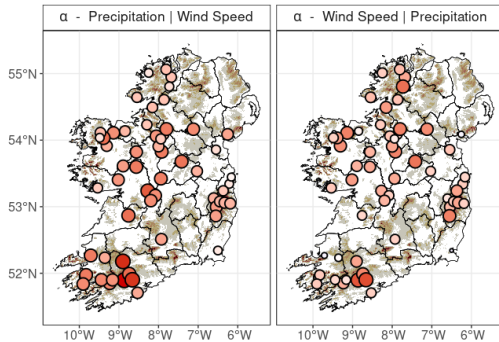


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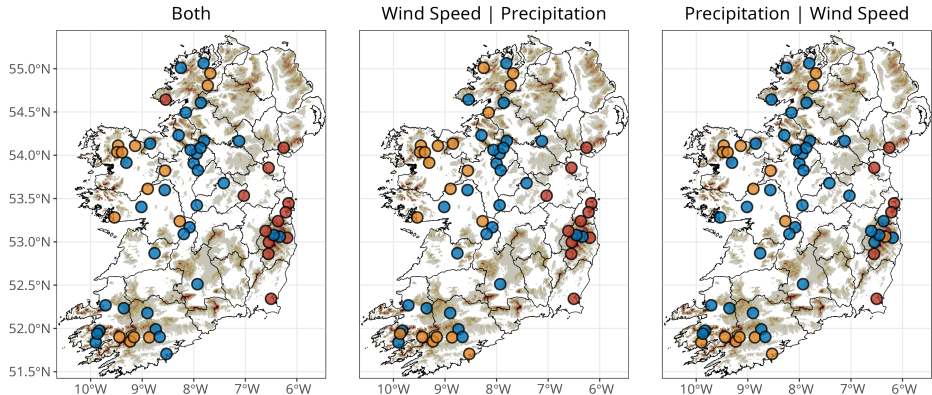


Ireland meteorological data

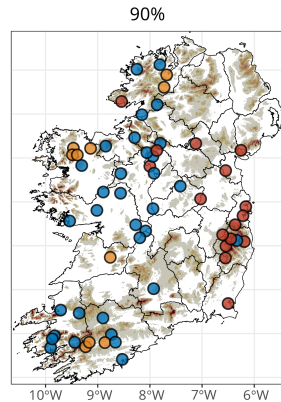
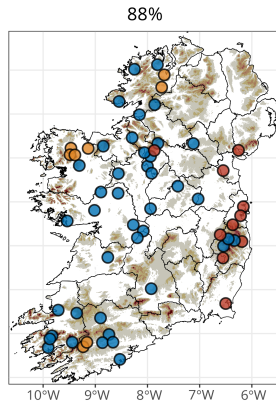
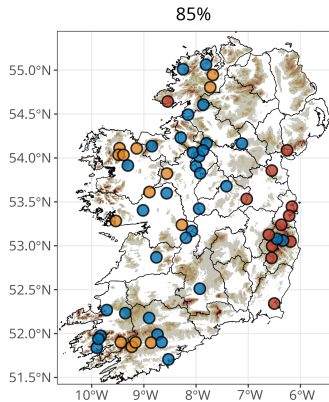
$\alpha_{i,s}, \beta_{i,s}$ estimates



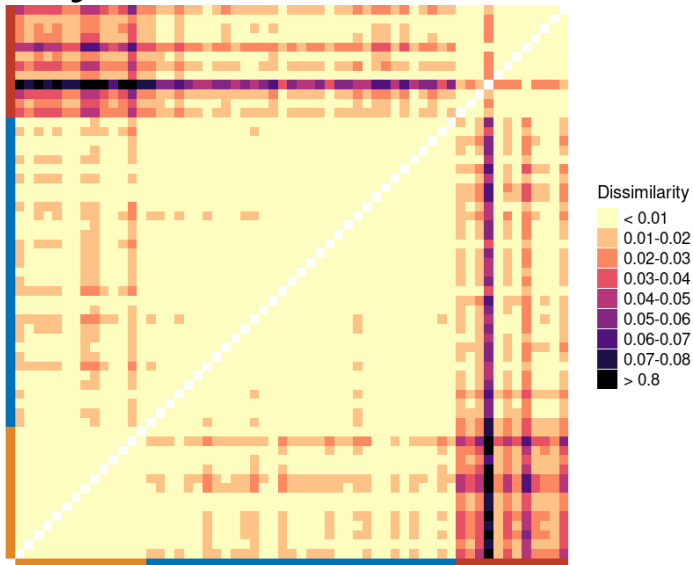
Ireland clustering



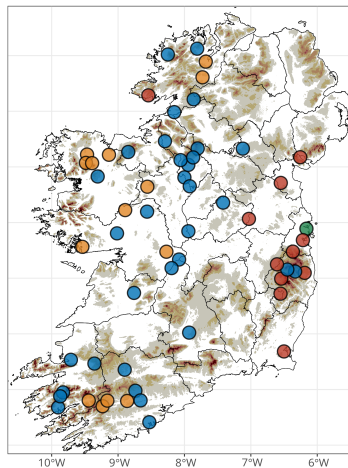
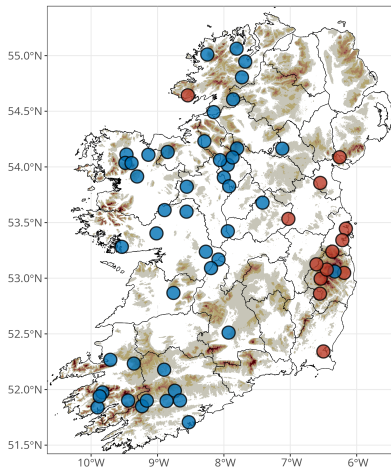
Sensitivity Analysis



Dissimilarity Matrix



$k = 2, k = 4$ **clusters**



Discussion

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- **Conclusion:**
 - Principled & effective clustering framework for CE models
 - Simulations & application show clustering can group sites with similar tail dependence, which may aid interpretation

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- **Conclusion:**
 - Principled & effective clustering framework for CE models
 - Simulations & application show clustering can group sites with similar tail dependence, which may aid interpretation
- **Limitations:**
 - **CE:** Gaussian assumption for $Z_{|i}$
 - **Clustering:** Uncertainty in CE fits is ignored in clustering

Thank You!



<https://arxiv.org/abs/2510.20424>



<https://github.com/potoole7/CeCl>

References I

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-  Heffernan, Janet E and Jonathan A Tawn (2004). “A conditional approach for multivariate extreme values (with discussion)”. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 66.3, pp. 497–546.

References II

-  Nielsen, Frank (May 2019). "On the Jensen-Shannon Symmetrization of Distances Relying on Abstract Means". In: *Entropy* 21.5, p. 485. ISSN: 1099-4300. DOI: 10.3390/e21050485. URL: <http://dx.doi.org/10.3390/e21050485>.
-  Vignotto, Edoardo, Sebastian Engelke, and Jakob Zscheischler (2021). "Clustering bivariate dependencies of compound precipitation and wind extremes over Great Britain and Ireland". In: *Weather and Climate Extremes* 32, p. 100318.

Appendix

$\text{JS}^{\text{G}_\lambda}$

- For $H \sim \text{MVN}(m, \Omega)$, $H^* \sim \text{MVN}(m^*, \Omega^*)$,

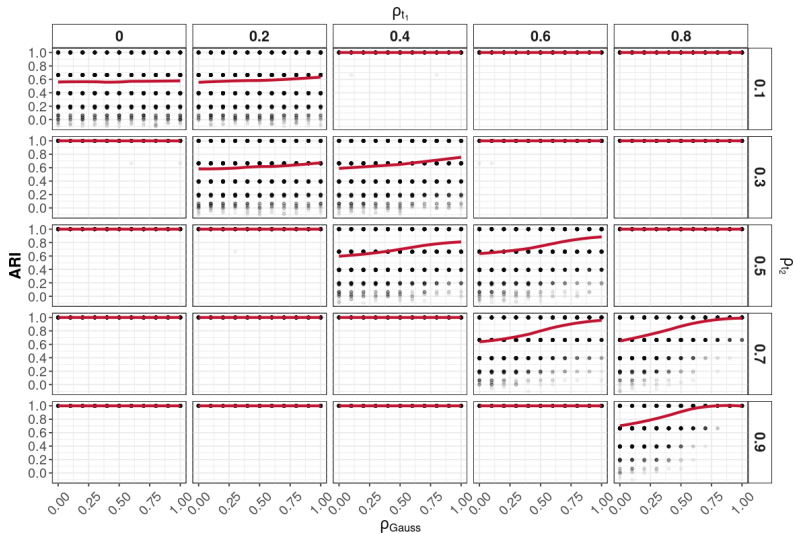
$$\begin{aligned} \text{JS}^{\text{G}_\lambda}(H \| H^*) &= \frac{1}{2} \left[(1 - \lambda) m^T \Omega^{-1} m + \lambda (m^*)^T (\Omega^*)^{-1} m^* \right. \\ &\quad \left. - m_\lambda^T \Omega_\lambda^{-1} m_\lambda + \log \left(\frac{|\Omega|^{1-\lambda} |\Omega^*|^\lambda}{|\Omega_\lambda|} \right) \right], \end{aligned}$$

$$m_\lambda = \Omega_\lambda \left[(1 - \lambda) \Omega^{-1} m + \lambda (\Omega^*)^{-1} m^* \right],$$

$$\Omega_\lambda = \left[(1 - \lambda) \Omega^{-1} + \lambda (\Omega^*)^{-1} \right]^{-1}.$$

- Ω_λ is matrix barycentre of Ω and Ω^*
- $\text{JS}^{\text{G}_\lambda}(H \| H^*) \leq \frac{1}{2} \max\{\lambda, 1 - \lambda\} [\text{KL}(H \| H^*) + \text{KL}(H^* \| H)]$

Mixture simulation (three variables)



More realistic example

